

# Blockbloom - GDG Winter Project 2024

## Mentors:

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## Project Info

### Week 1: Understanding Blockchain Basics

- **Evolution of the Web**

Gain an understanding of how the internet evolved from static Web 1.0 to dynamic Web 2.0, and its transition toward decentralized Web 3.0. What problems does Web 3.0 aim to solve? How does blockchain technology fit into this? ([Link](#))

- **"But how does Bitcoin actually work?"**

Watch this [video](#) to get a beginner-friendly introduction to the technical underpinnings of Bitcoin and blockchain. Concepts covered include transactions, blocks, and distributed ledgers.

- **Cryptographic Hash Function ([Link](#))**

Understand what cryptographic hash functions are and their importance in securing data. Study characteristics such as determinism, collision resistance, and pre-image resistance.

Learn about [SHA256](#), the cryptographic hash function widely used in blockchain. Explore its role in generating unique digital signatures for data blocks.

- **What is a Blockchain? ([Link](#))**

Understand the core structure of blockchain: blocks, chains, and nodes. Key questions to consider: How do blockchains achieve immutability? Why is decentralization important?

Study the mechanism behind [PoW](#) and its role in ensuring consensus in blockchain networks. Key aspects to learn: mining, computational puzzles, and energy consumption. Explore [PoS](#) as an alternative consensus mechanism. Learn about staking, validators, and how PoS addresses some limitations of PoW. Engage with this interactive [demo](#) tool to visualize how Proof of Work operates.

- **51% Attack**

Understand what a [51% attack](#) is and why it poses a significant risk to blockchain networks. Study scenarios where this attack becomes feasible and measures to prevent it.

- **Bitcoin whitepaper**

Read and analyze the original [Bitcoin whitepaper](#) by Satoshi Nakamoto.  
Key focus: Peer-to-peer transactions, cryptographic proof, and decentralized trust.

- (Optional) [Zero knowledge proofs](#)

Slides: [Lecture 1](#)

Assignment 1: [Assn1](#) (Deadline: 22 December 2024)

## Week 2: Introduction to Ethereum

### Ethereum Basics:

#### Topics:

1. **Overview of [Ethereum \(Whitepaper\)](#):** Detailed study of Ethereum's whitepaper and key concepts, highlighting how it differs from Bitcoin in terms of purpose and functionality.
2. **Use Cases of Ethereum:** Exploration of Ethereum's applications beyond cryptocurrency, including smart contracts, decentralized applications (dApps), and DeFi.

### Getting Hands Dirty:

#### Activities:

1. **[Session Summary](#):** A recap of the practical demonstration conducted during the session, covering key steps and outcomes.
2. **Required Tools Usage:**
  - [Metamask Installation](#): Step-by-step guide to setting up the Metamask browser extension for interacting with Ethereum.
  - [Sepolia Faucet](#): Learn to obtain test Ether for Sepolia, a test network.
  - [OpenZeppelin Contract Wizard](#): Utilize OpenZeppelin's tool to generate secure smart contract templates.
  - [Remix IDE](#): Hands-on practice with Remix, an online IDE for developing and deploying smart contracts.
  - [Pinata](#): Understanding how to use Pinata for IPFS-based file storage and NFT metadata hosting.
  - [OpenSea Testnets](#): Explore OpenSea's testnets for NFT creation, minting, and listing.
  - [Sepolia Etherscan](#): Familiarize yourself with Sepolia's blockchain explorer to monitor transactions and deployed contracts.

Slides: [Lecture 2](#)

Assignment 2: [Ass 2](#) (Deadline: 30 December 2024)

## Week 3: Solidity

Solidity [documentation](#) : A deep dive into the official Solidity documentation to understand the syntax, features, and best practices for writing smart contracts.

[Code](#) (Discussed in the meet) : Review and explanation of the Solidity code examples discussed during the session, including their purpose and functionality.

Assignment 3: [Ass 3](#) (Deadline: 8 January 2025)

## Week 4: DApp development

Step-by-step guidance on deploying a decentralized application (DApp) to a blockchain network, including initial setup and advanced deployment techniques. Following are the videos recorded during the session.

- [Deployment part-1](#)
- [Deployment part-2](#)

Learn how to interact with deployed smart contracts using tools like Metamask, web3.js, or ethers.js, covering both basic and advanced functionalities. Following are the videos recorded during the session.

- [Interaction part-1](#)
- [Interaction part-2](#)

Slides: [Lecture 3](#)

[Code used in the videos](#)

**[ReactJS](#) Integration:** Understanding how to use ReactJS for building user interfaces for DApps, including connecting with smart contracts and blockchain.

Final project submission guidelines: [Here](#)